Ericsson PE Node IND-PE-03 Indiranagar Exchange QoS Congestion Event SOP

Node: IND-PE-03 | **Vendor:** Ericsson | **Severity:** Minor | **Alarm Code:** ALARM_2794

Description

The QoS congestion event on Ericsson's PE node IND-PE-03 at the Indiranagar Exchange is primarily caused by the occurrence of microbursts, interface tail drops, and QoS queue starvation. This is usually a result of the node's inability to handle bursty traffic efficiently, which can be attributed to the node's limited buffer sizes, insufficient policing and shaping mechanisms, or misconfigured QoS policies. Furthermore, the high-speed interfaces and dense wavelength division multiplexing (DWDM) technology employed in modern networks have increased the likelihood of congestion events due to the amplification of packet bursts. Additionally, the complex interaction between multiple protocols and services, such as MPLS, IP, and Ethernet, can lead to queue starvation and other congestion-related issues. Therefore, it is crucial to identify and address the root cause of the congestion event to prevent service degradation and ensure optimal network performance.

Pre-requisite Checks

- Ensure that remote hands are dispatched to the Indiranagar Exchange for physical access to the equipment.

- Verify the status of the redundant link to ensure it is operational and not contributing to the congestion event.

- Check the environmental conditions, such as temperature and humidity, to rule out any equipment-related issues.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Insufficient policing and shaping mechanisms	Medium
Limited buffer sizes	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy, Expected Output: Policy name: Priority: DSCP: , Description: This command displays the current QoS policy configuration, allowing us to verify if the policy is properly configured and if it is being applied to the affected interface.

Command: show interface statistics, Expected Output: Interface: In-Bytes: In-Packets: Out-Bytes: Out-Packets: , Description: This command provides detailed statistics for the affected interface, enabling us to identify any anomalies in traffic patterns or errors
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that may be contributing to the congestion event.

Command: show policing statistics, Expected Output: Policing name:
Packets: Bytes: Exceeds: , Description: This command displays policing
statistics for the affected interface, allowing us to determine if the
policing mechanism is functioning correctly and if it is enforcing the
intended policy.
```

Resolution Steps

Step 1: Step: Verify and correct QoS policy configuration, Command: configure qos policy priority dscp , Description: This step involves verifying the current QoS policy configuration and correcting any misconfigurations to ensure that the policy is properly applied to the affected interface.

Step 2: Step: Implement policing and shaping mechanisms, Command: configure policing , Description: This step involves implementing policing and shaping mechanisms to control bursty traffic and prevent congestion events.

Step 3: Step: Increase buffer sizes, Command: configure interface buffer-size , Description: This step involves increasing the buffer sizes for the affected interface to prevent queue starvation and congestion events.

Post-Check Verification

```
Command: show qos policy, Expected Output: Policy name: Priority:
DSCP: , Description: This command verifies that the corrected QoS
policy configuration is being applied to the affected interface.

Command: show interface statistics, Expected Output: Interface:
In-Bytes: In-Packets: Out-Bytes: Out-Packets: , Description: This
command verifies that the corrected policing and shaping mechanisms
are functioning correctly and preventing congestion events.

Command: show policing statistics, Expected Output: Policing name:
Packets: Bytes: Exceeds: , Description: This command verifies that the
increased buffer sizes are preventing queue starvation and congestion
events.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: configure qos policy priority dscp , Description: This step involves restoring the original QoS policy configuration to its previous state in case the correction made in step 1 fails to resolve the congestion event.

- Step: Disable policing and shaping mechanisms, Command: configure policing , Description: This step involves disabling the policing and shaping mechanisms implemented in step 2 to prevent any adverse effects on network performance.

- Step: Restore original buffer sizes, Command: configure interface buffer-size , Description: This step involves restoring the original buffer sizes to prevent any adverse effects on network performance.